Diseases of bovine hoof

Kevin Kai-Chen Hsu, BVM, MS

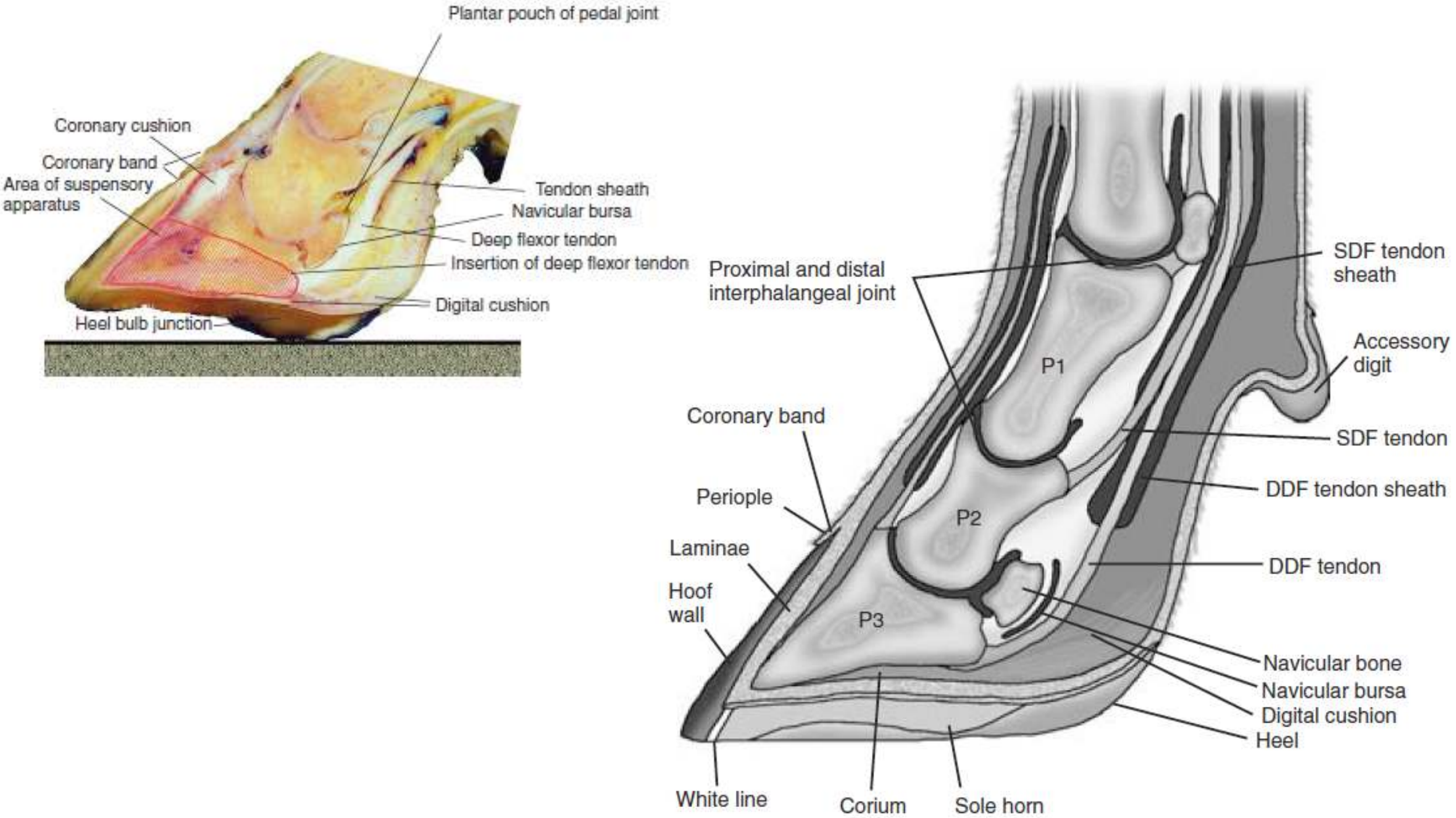


Figure 13.3 Functional anatomy of the bovine foot: SDF, superficial disital flexor; DDF, deep digital flexor.

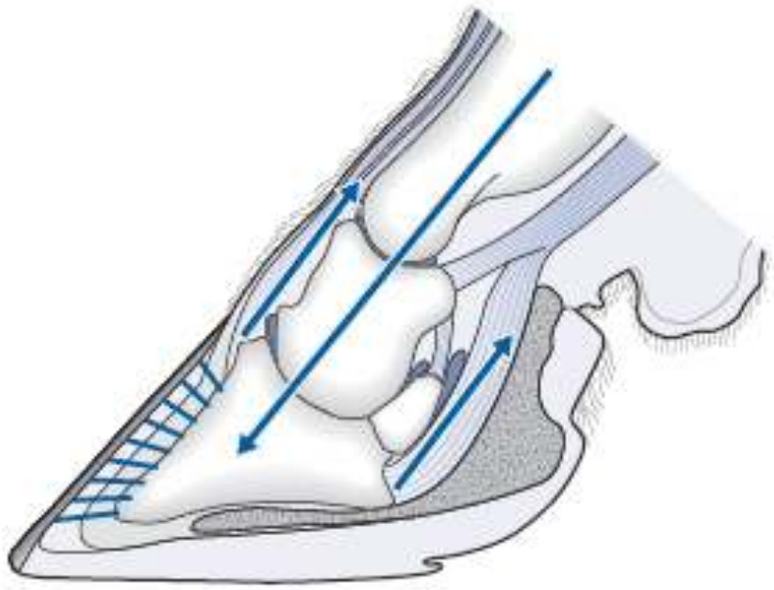




Figure 2-17 The appearance of the normal laminae (dermal lamellae) of an exungulated claw. (Courtesy of P Ossent)

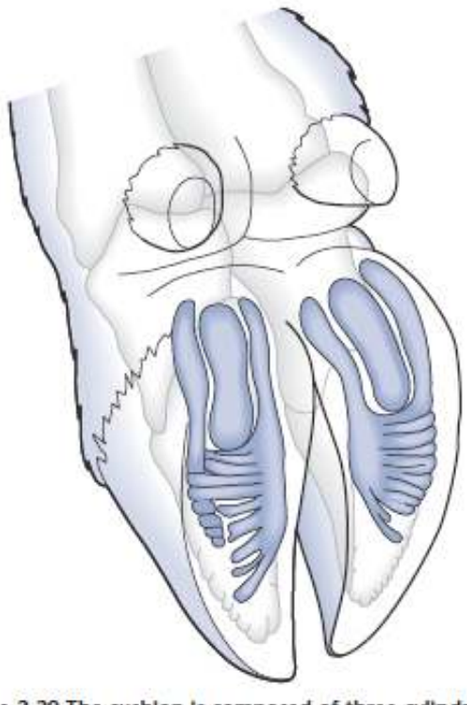
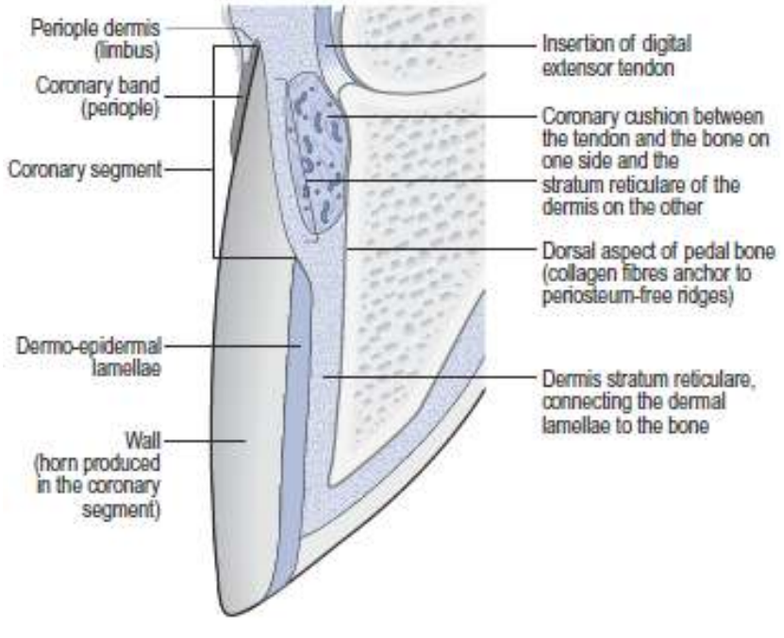


Figure 2-29 The cushion is composed of three cylindrical parallel-oriented bodies each with a capsule of connective tissue filled with soft fat. Towards the tip of the pedal bone a system of thin transverse oriented fat strands is present in continuity with the fat cylinders. (Courtesy of C Lischer & P Ossent)



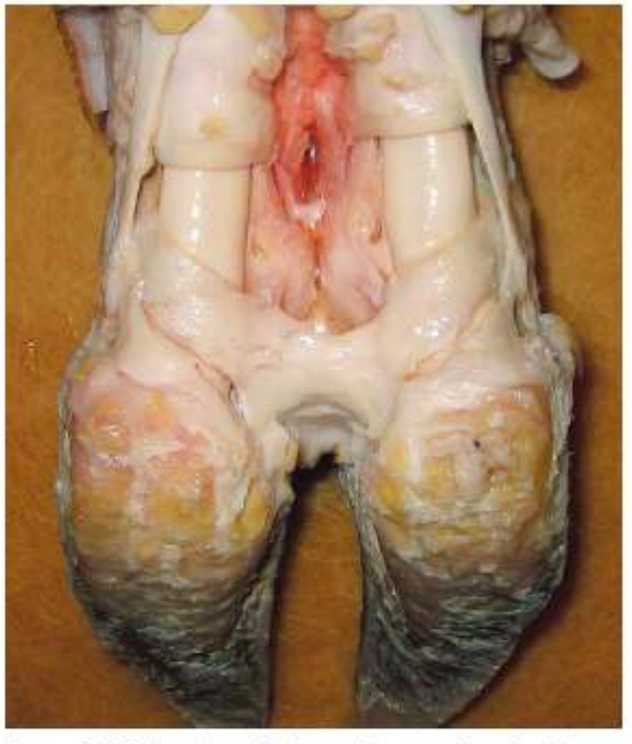


Figure 2-27 Dissection of the cruciate complex showing its relationship to the digital retinaculum. (Courtesy of C K W Mülling)

Biomechanics

Biomechanics of the cow shows that the front feet bear weight in a more flexible manner than the hind feet. Also the forefeet support heavier weights than the hind feet. When studying the movements of the hind feet, a lot of weight appears to be shifting from one foot to the other, even when the cow is standing still. A weighing test with scales proves that this is the case. It would not be a problem if the shifting weight was spread evenly on both claws of the foot. The truth is that the outer claw takes more weight than the inner claw.

If we compare the size and the weight-bearing surface, the outer claw is larger and its weight-bearing surface is flatter. The sole of the inner claw slopes at a sharper angle and provides little stability, especially on a hard surface. The outer claw must compensate for this by taking more weight. Consequently, it will wear thin sooner than the inner claw. As a reaction the corium (living tissue) produces extra horn which, in the long run, will result in the outer claw being overburdened.

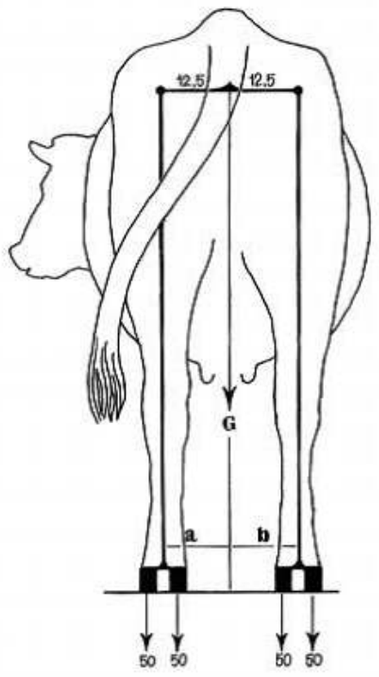


Fig 1: A cow standing still has the weight distributed equally between hind legs and hooves.

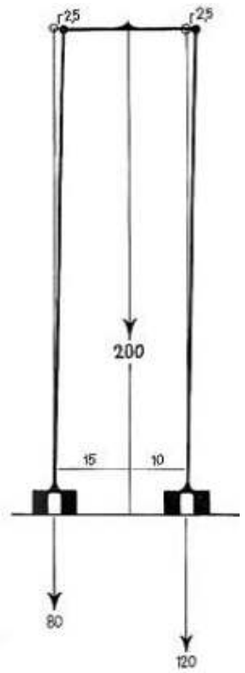


Fig 2: When the cow leans to one side more weight goes on the outside leg

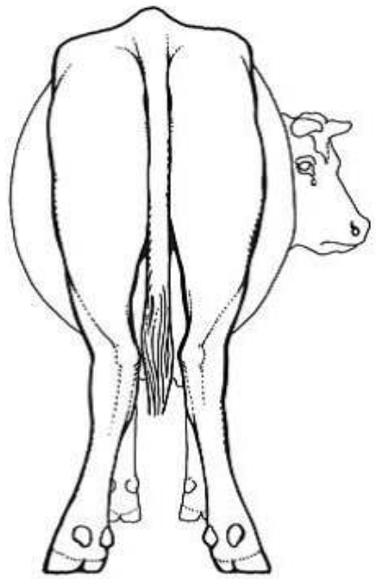


Fig 3: When there is overload the cow adopts a cow hock stance

Inside the foot the last bone, the pedal bone, takes and transfers the weight of the cow to the corium and then to the sole. Between the pedal bone and the corium a digital cushion functions as a shock absorber. Normally the shock is absorbed sufficiently, but when soles are very thick the cushion cannot absorb the shock properly. The pedal bone hits the corium often and hard so that capillaries burst and a bruising (blood stain) shows in the sole. If this goes on the corium will get so badly damaged that it no longer forms horn on that spot and in the sole a hole appears, called a sole ulcer. Sole ulcers are also called typical sole ulcers. The part of the pedal bone causing this ulcer shows a small lump at that spot. This causes the (typical) ulcer. By trimming back the sole of the outer claw to its normal thickness sole ulcers can be prevented.

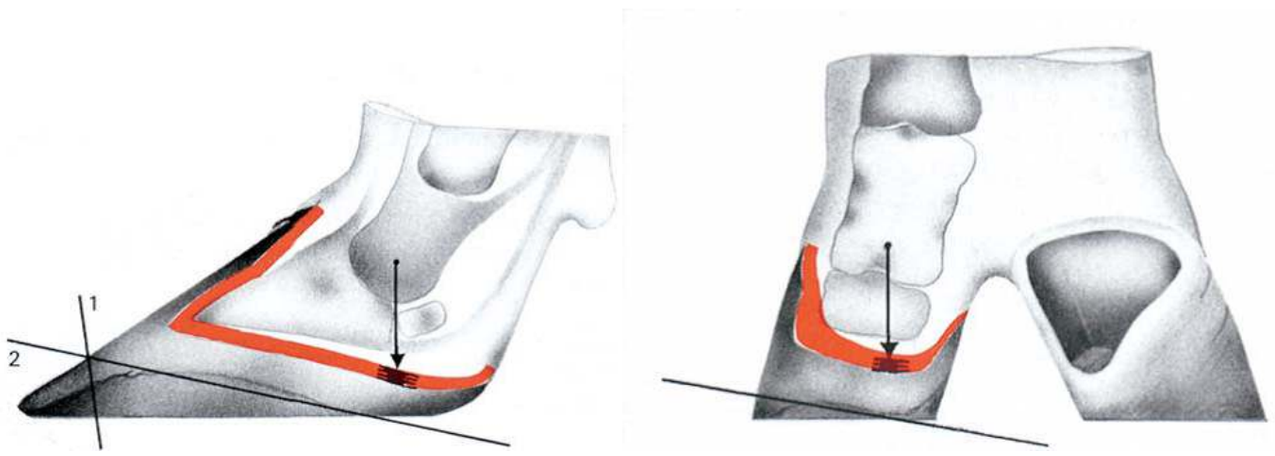
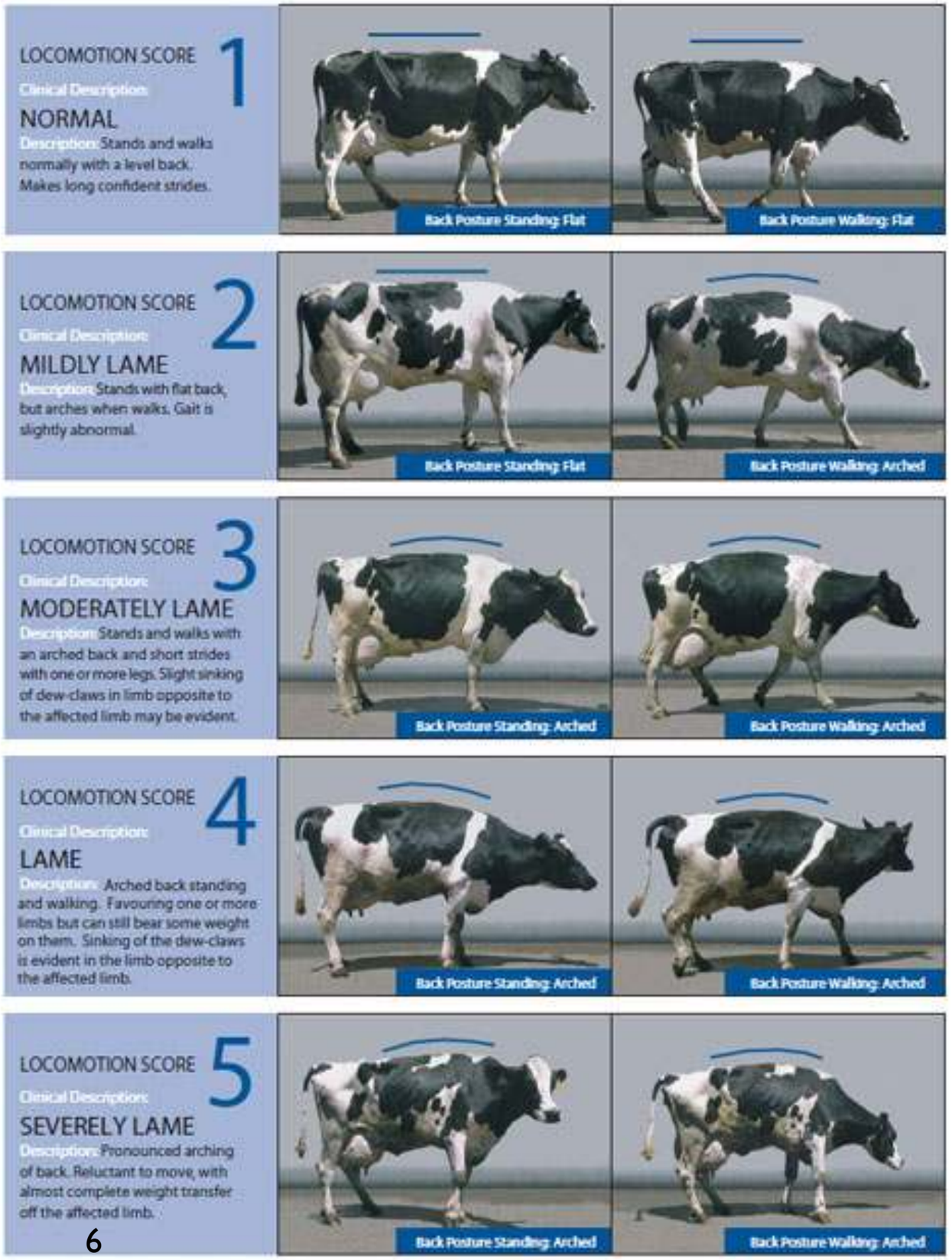




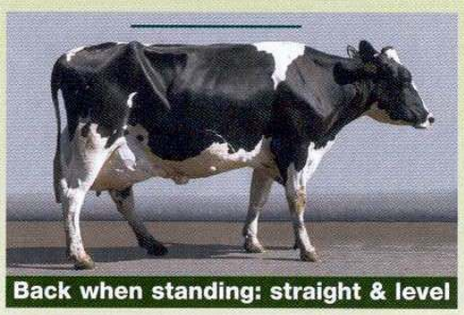


Figure 4 and 5 *Typical sole-lesion starts after long pressure on the outer claw. This can form the typical sole ulcer. Overbudening puts pressure on this spot* .1 and 2 depicts trimming areas to t*ake away the pressure so that the bodyweight shifts to the front.*

Locomotion Scoring for Dairy Cattle



* Adapted from Sprecher, D.J., Hostetler, D.E., Kanawha, J.B. 1997. Theriogenology 47:1179-1187 and contribution from Cook, N.B. University of Wisconsin.


Back when standing: straight & level

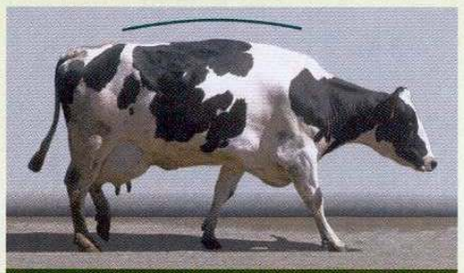

Back when standing: straight & level

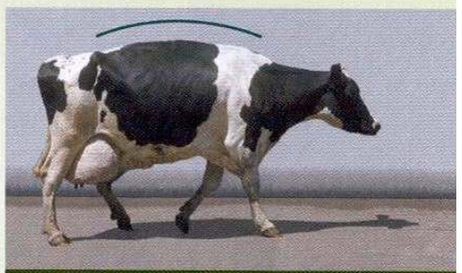

Back when standing: arched


Back when standing: arched

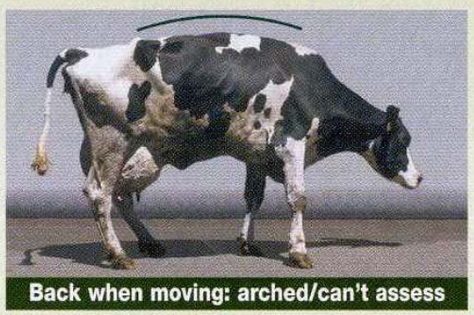

Back when standing: arched


Back when moving: straight & level


Back when moving: arched


Back when moving: arched


Back when moving: arched


Back when moving: arched/can't assess

Locomotion score 1
The healthy walk
The animal stands and walks normally. She places all hooves with confidence; the rear hooves are placed where the front hooves were.
Conclusion: good.*

Locomotion score 2
The slightly abnormal walk
The cow stands normally but arches her back when she starts to walk. Her head is held lower and extended from the body. Her gait is slightly abnormal.
Conclusion: this cow needs attention.*

Locomotion score 3
The lame cow
The cow arches her back both when standing still and when walking. She takes short steps with one or more legs.

Conclusion: this cow requires treatment today.*

Locomotion score 4
The very lame cow
The animal tries to reduce the weight on one or more feet. She arches her back both when standing still and when moving.

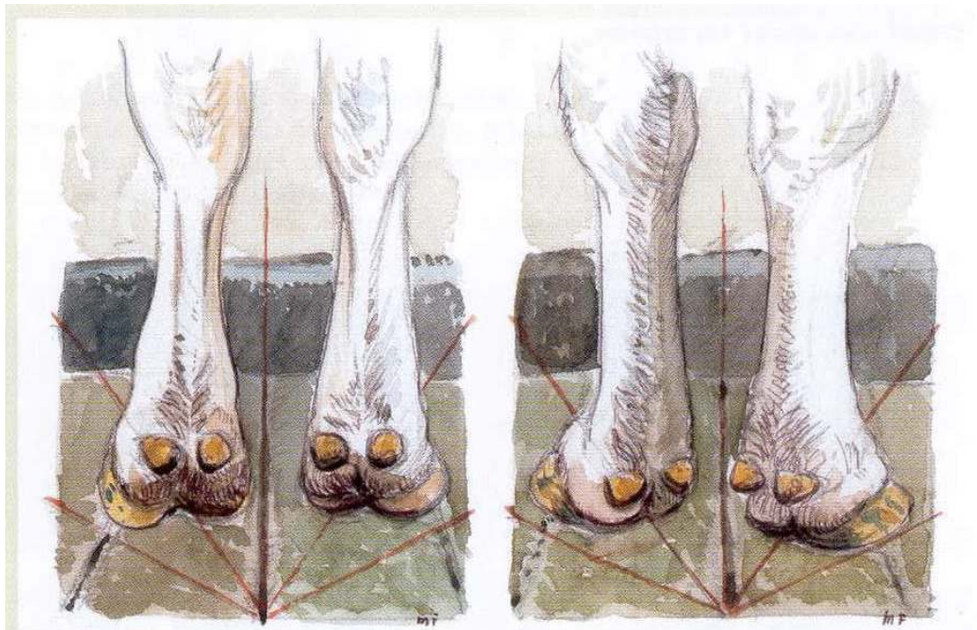
Conclusion: this cow needs immediate treatment and nursing care.*

Locomotion score 5
The crippled cow
The animal has an arched back. She refuses to stand on a particular hoof, or limps. She prefers to remain lying down, or has great difficulty when trying to get up.
Conclusion: this very lame, sick cow needs intensive care and professional treatment.*

Source: Steven L. Berry, DVM, MPVM
Univ. of Davis, CA, and Zinpro® Corporation (1997)

*These are my conclusions. J.H.

Leg Score



Leg score

The leg score gives an impression of the stance of the hind legs and can be used to manage foot problems. It is related to the height difference between the inner and outer claw and the way the cow places her foot. Cows rotate their feet outwards to relieve painful areas of the sole and are more likely to do this on slippery floors, when they walk with more weight on their heels.

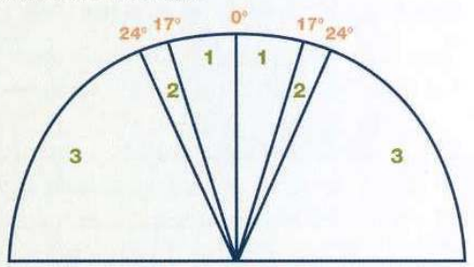
Leg score 1: the ideal situation; but infections or haemorrhages can still occur.

If there is a high proportion of leg scores 2 and 3, you should look at the technique and frequency of foot trimming, and other factors affecting stance.

Leg score 2: the placement of the hooves could be better. Suggests a high incidence of foul-in-the-foot and/or a need for regular, corrective foot trimming.

Leg score 3: undesirable situation. The cow or cows need immediate attention (trim hooves and treat).

(From J. van Amerongen)



Leg score calculator

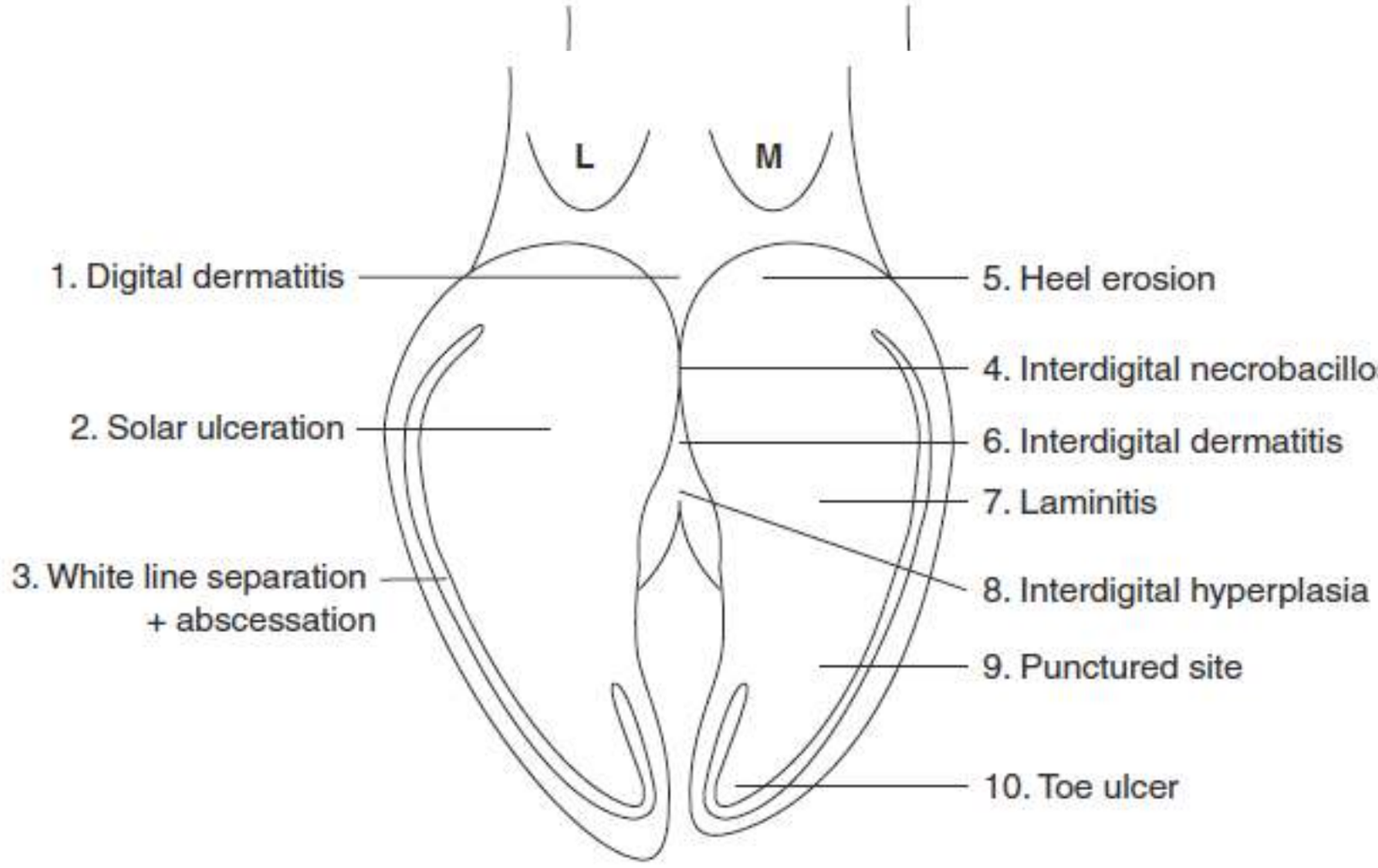
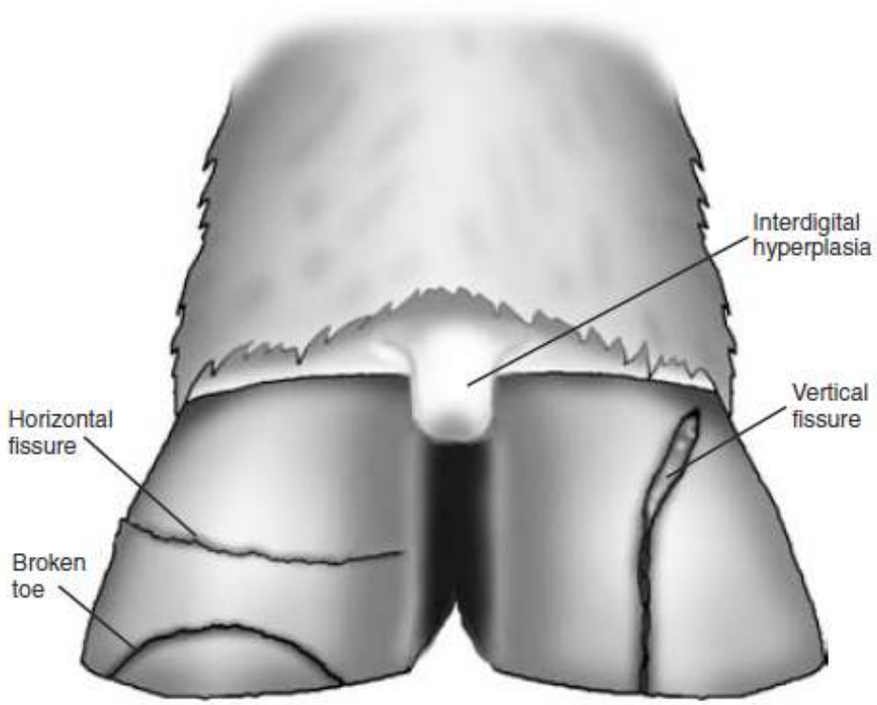


Figure 7.2 Important conditions of the bovine digit.
1. digital dermatitis; 2. sole ulceration; 3. white line separation and abscessation; 4. interdigital necrobacillosis; 5. heel erosion; 6. interdigital dermatitis; 7. laminitis/sole haemorrhage; 8. interdigital hyperplasia; 9. punctured sole; 10. toe ulcer.

Close inspection of the standing animal

Horizontal and vertical fissures of the hoof wall
a horizontal fissure that has progressed to the toe may cause a piece of the toe horn to snap off prematurely resulting in a‘broken toe’



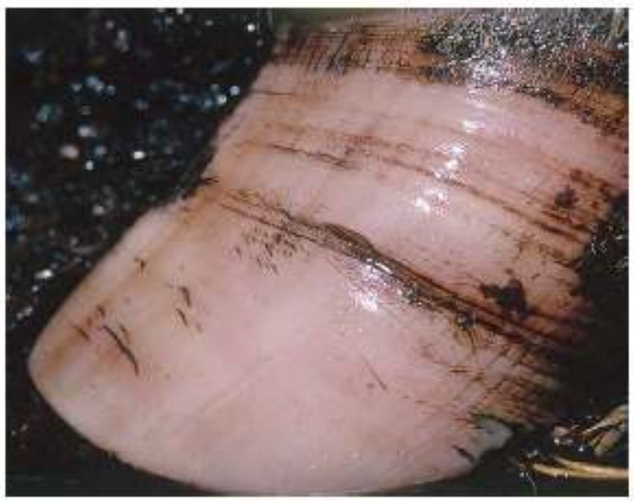


Figure 4-5 The simplest groove is often referred to as a ‘hardship groove.’ A groove such as this might form as the result of a sudden change in diet or a febrile disease such as metritis.



Figure 4-7 The mark furthest from the skin/horn junction represents the time at which the animal calved. The groove closest to the skin/horn junction was caused by a sudden change in diet.

Close inspection of the standing animal

Coronary band lesions
- may be caused by digital dermatitis or trauma

Severe swelling of the area proximal to the coronary band
- may indicate a septic arthritis of the distal interphalangeal joint, foul or ascending infection from white line disease.

Coronary band vesicles/erosions
- if foot-and-mouth disease is suspected, a careful examination for vesicles or ruptured vesicles is required; if mucosal disease is suspected, erosions may be present.

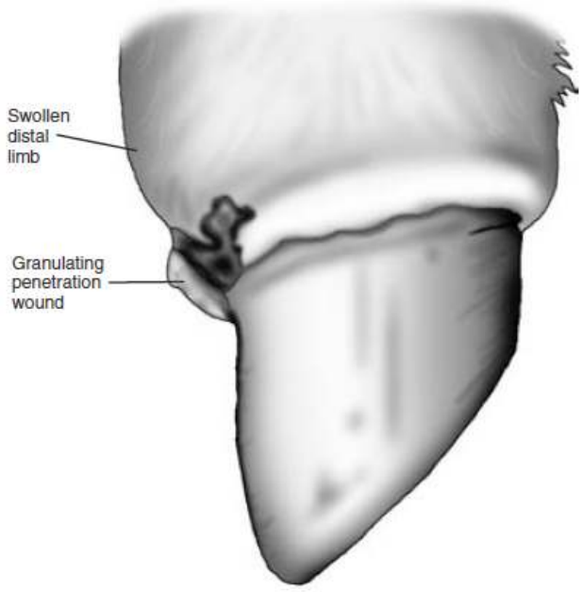


Figure 13.5 Septic arthritis of the distal interphalangeal joint.

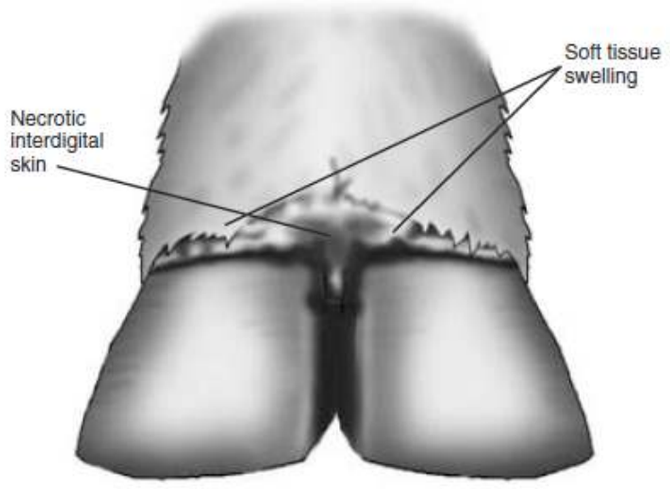


Figure 13.6 Foul in the foot.

Close inspection of the standing animal

Grossly overgrown and corkscrew claws
will be self evident.

Changes in claw conformation
may be seen in chronic laminitis and corkscrew claw. There may be horizontal ridges extending the length of the claw with a concavity of the wall giving the Turkish slipper appearance. Corkscrew claw presents as a severe rotation of the hind-leg lateral claw.

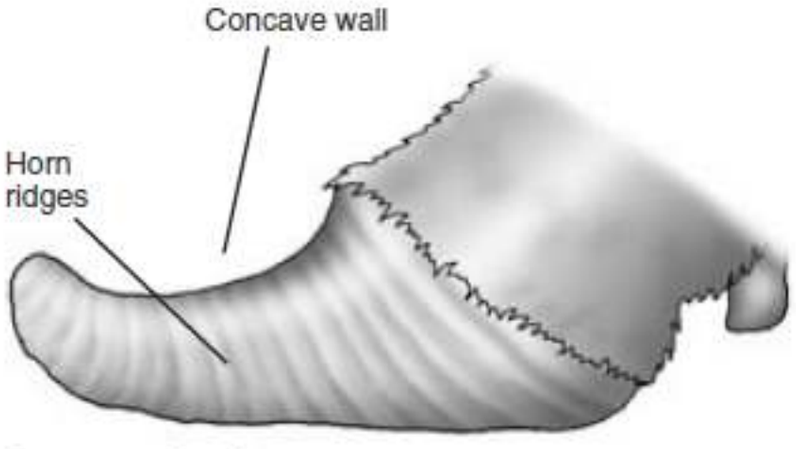


Figure 13.7 Chronic laminitis.

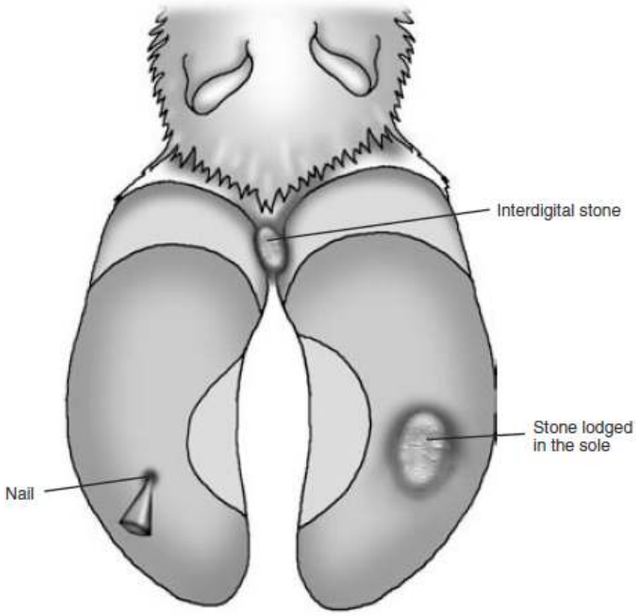
Figure 4-20 This is a typical slipper foot showing a flattened concave dorsal surface.

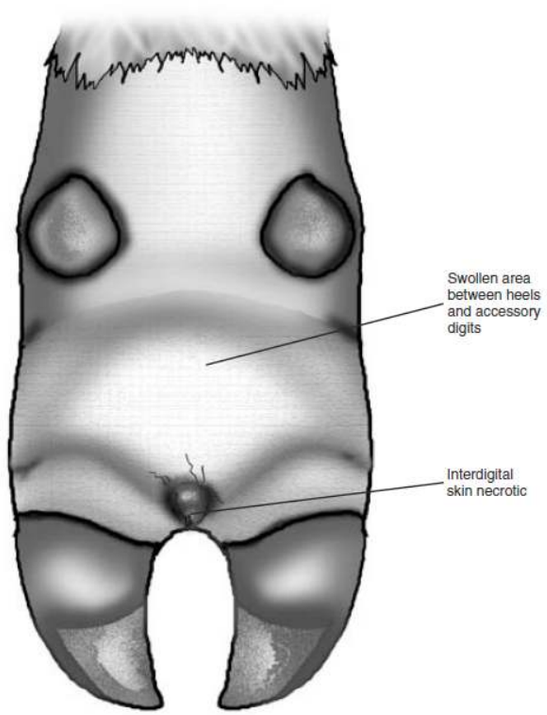


Figure 4-21 These claws have square toes, the dorsal surface is flat and concave. The coronary band is rough and has a horny fringe.

Examination of the foot

The most common causes of lameness in cattle are:
foul in the foot, white line disease, sole ulceration, penetration of the hoof by sharp objects, heel erosion, digital dermatitis and laminitis.







Swelling of the skin around the distal limb

Foul presents as a swelling of skin between heels and accessory digits caused by inflammation of the skin and subcutaneous tissue. The claws may be forced apart with necrotic interdigital skin. Infection of the synovial structures of the distal limb, including the distal and proximal interphalangeal joints, results in severe generalised swelling of the distal limb.

Examination of the sole

Laminitis
This causes pressure pain and loss of‘adhesion’of the horn to the underlying structures of the foot. The blood supply to the structures supporting and producing the horn of the hoof is compromised, and poor quality soft horn may result. Laminitis may predispose the foot to sole ulceration, white line disease, rotation and penetration of the sole by P3. Adouble sole may follow a bout of laminitis. Laminitis presents with a yellow exudative/ haemorrhagic discolouration of the solar horn.

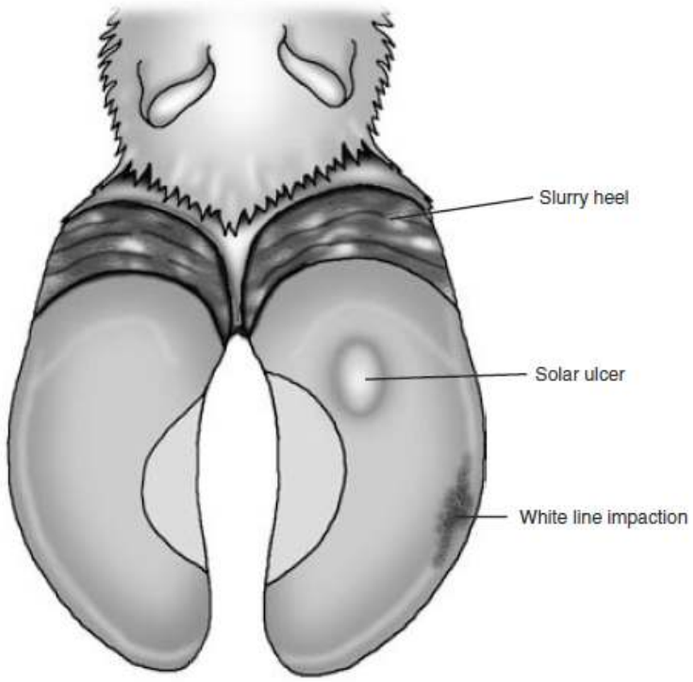




Figure 4-9 These are very bruised soles. (Courtesy of J Malmo)



Figure 4-25 Congestion, edema, and hemorrhage of the laminae are evidence of sinkage. Usually, it takes some time for blood to discolor the 'lamellar' region; this gives some indication of the time elapsed since the onset of pathological changes in the live animal. (Courtesy of P Ossent)



Figure 4-24 The digital region of a cadaver is immersed in water at 60°C for 20 minutes. The capsule is fixed in a vice and the digit levered away. (Courtesy of P Ossent)

Examination of the sole

Classical solar ulcers

These are recognised as circumscribed areas where horn is missing or the horn is of very poor quality. The corium is exposed or has been protected by the production of granulation tissue. The ulcers occur towards the back of the sole, two-thirds of the distance between the toe and the heel directly over the posterior border of the pedal bone. They are usually the size of a bottle top. Less commonly solar heel ulcers may be present which are set further back on the sole. Ulcers are often bloody due to presence of granulation tissue, and very painful.

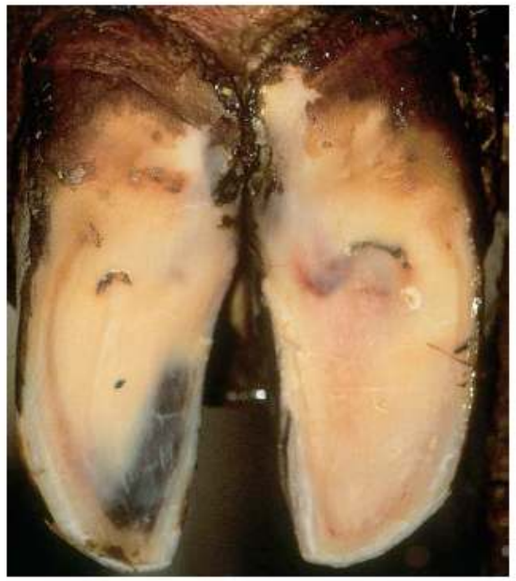


Figure 7-2 The first sign of a sole ulcer may be a hemorrhage over the typical site.



Figure 7-3 Sometimes an ulcer is not discovered until superficial layers of sole horn are pared away. (Courtesy of C Bergsten)



Figure 7-4 If sole ulcers are neglected, infection will cause damage to the flexor tendon, spread up the tendon sheath, and/or cause serious infection of the pedal joint. (Courtesy of C Bergsten)

Examination of the sole

White line disease

The white line is a point of weakness, and dirt may become impacted and provide a portal of entry for infection if the integrity of the join between the horn of the wall and the horn of the sole is compromised. Once the claw has been trimmed affected regions of the white line can be readily identified as distinct black thick zones.



Figure 7-12 The earliest sign is a distinct hemorrhage in the white line. (Courtesy of J Naylor)



Figure 7-14 Once superficial horn is removed pus is released. If the lesion is not detected early a destructive retroarticular abscess will be the likely result. (Courtesy of J Malmo)

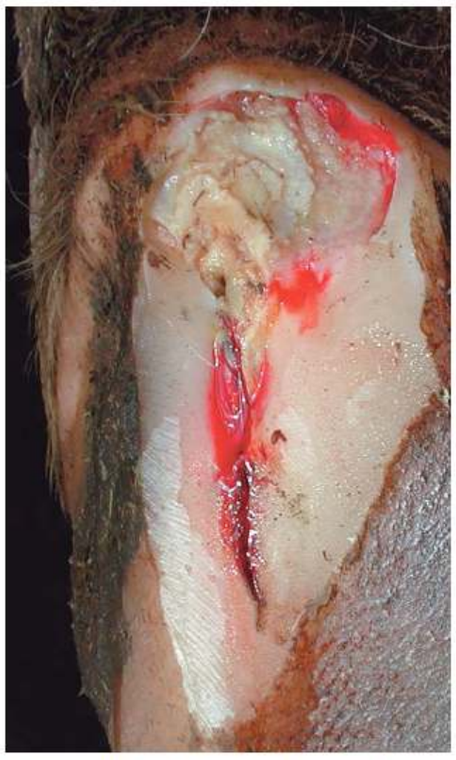


Figure 7-15 The same lesion as shown in Figures 7-13 and 7-14 with the extent of the lesion exposed. Note that the pus is extending to and escaping from the skin/horn junction above the heel bulb. (Courtesy of J Malmo)



Figure 7-16 This case is similar to the last except that the lesion is now 'open' and starting to accumulate foreign bodies. (Courtesy of J Malmo)

Solar abscess

White line disease can result in ascending infections or lateral infections. The latter produce solar abscesses. These may be detected during corrective trimming by a sudden thinning of the sole followed by the appearance and drainage of pus. Alternatively, it may present as a painful pliable thin sole at the end of Stage 1b of the foot trimming process. Pain testers can be useful to localize the area of pain, and further thinning of the sole may puncture the abscess. Occasionally, on applying pressure, pus can be seen to exude from the original portal of entry in the white line.

Examination of the sole



Figure 7-24 In this picture the track from the white line to the coronary band has caused considerable damage. (Courtesy of P Ossent)



Figure 7-25 Infection may be forced along the spaces between the lamellae (p. 89) to form a track. An elliptical incision has been made to remove wall to expose tracks. This procedure has to be considered as a very important measure not only in the treatment but also in the prevention of the complication of white line (closed) defects. Lacking the side wall, penetration of foreign objects in the white line defects, and consequent development of an abscess, will become almost impossible. (Courtesy of A Brizzi)



Figure 7-26 In this case an elliptical cut has been made, exposing a more marked destruction of the dermal/epidermal junction. (Courtesy of A Brizzi)

Examination of the sole



Figure 7-10 Heel erosion can be a contributing factor. Loss of heel horn causes weight to be shifted forward to the typical site of a sole ulcer.

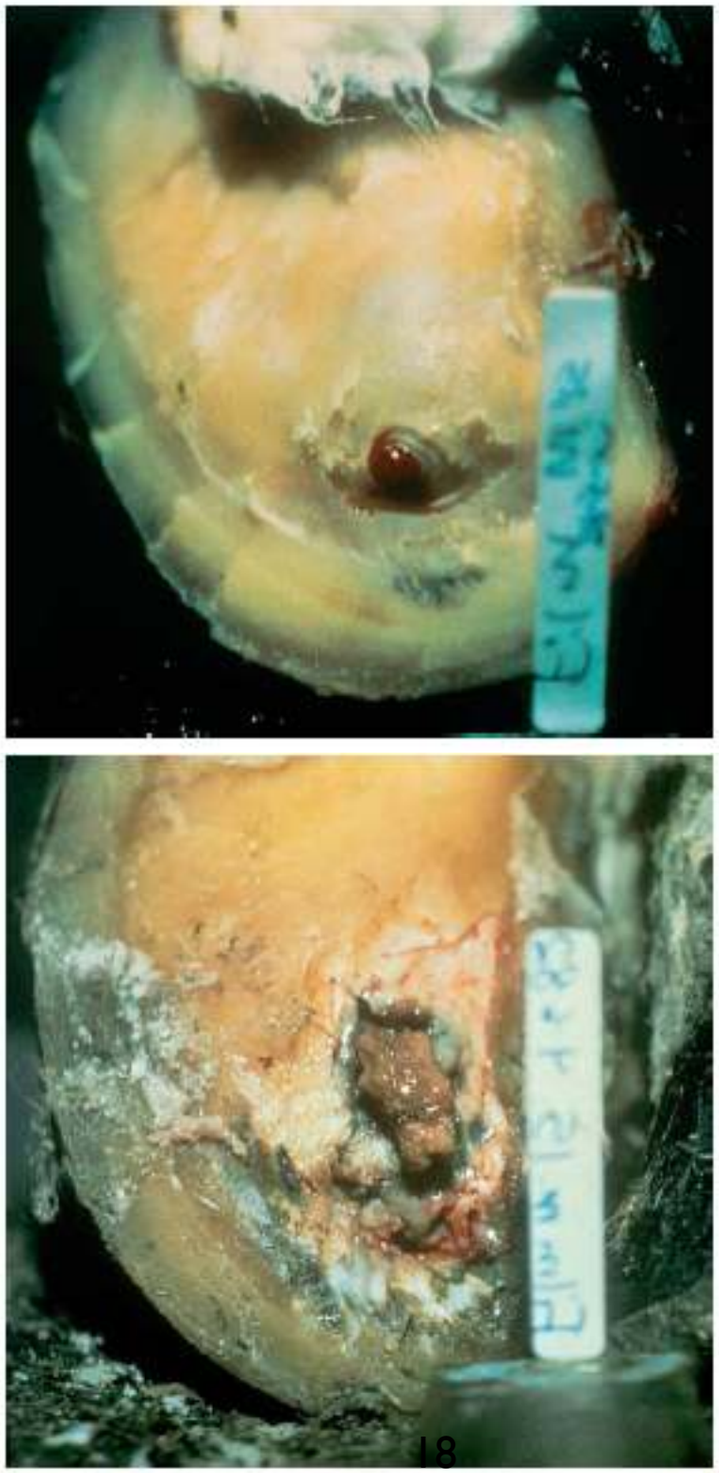






Figure 7-36 Careful dissection of the sole revealed the extent of the abscess. Although the pedal bone is exposed the dermis will regenerate if the lesion is adequately protected from further damage and infection. (Courtesy of J Malmo)

Examination of the sole

Lesions caused by digital dematitis are commonly found on the skin between the bulbs of the heels and are recognised as a raw shiny area of skin with the shape and colour of a strawberry cut in half (‘strawberry heel’). The lesion is very painful when touched. In the chronic form known as verrucose dematitis, a plume of fulminating keratin ‘threads’ is seen.

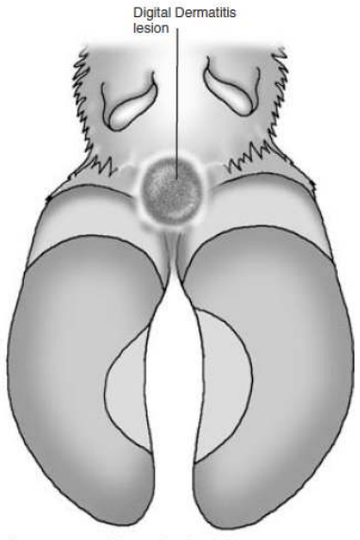


Figure 13.11 Digital dermatitis (strawberry heel).

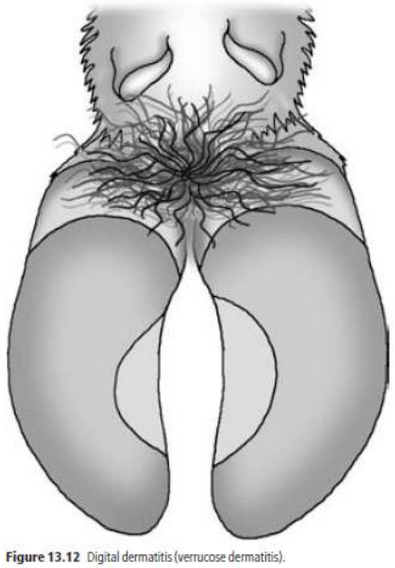


Figure 13.12 Digital dermatitis (verrucose dermatitis).





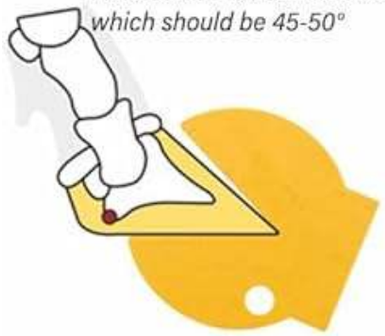


Hoof Trimming

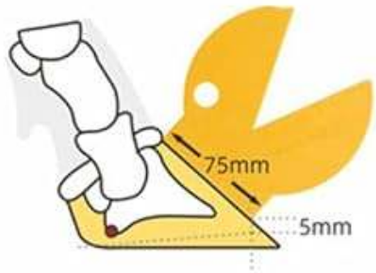
Dutch 5 steps method

Functional Trimming
1) Trim toe length to correct length
2) Match untrimmed claw length/level
3) Model (dish) out sole ulcer site
Corrective (Therapeutic) Trimming
4) Relieve weight off painful claw
5) Remove loose/under-run horn and hard ridges in high risk zones

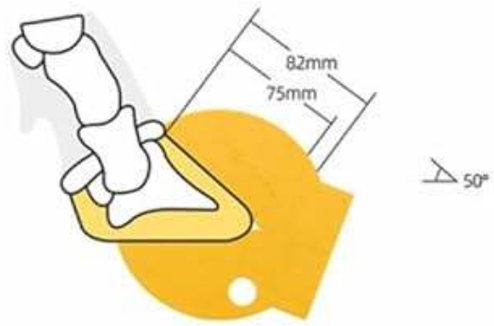
① *Before trimming the hooves, check the angle between the bottom and wall of the hooves, which should be 45-50°*



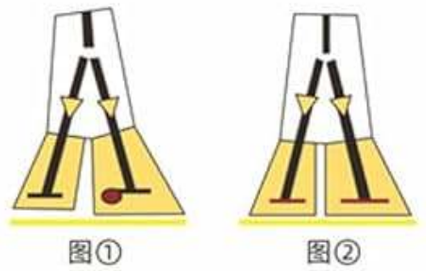
② *After trimming the hooves, the hoof wall should be 7.5cm long and the toe of the hoof bottom should be 5cm thick*



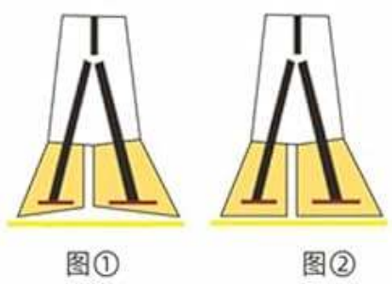
③ *Measurement of size and angle of the hoof wall after hoof repair*



④ *Before trimming, the height of the inner and outer toes is different, after trimming, they are on the same horizontal line*



⑤ *Before trimming, there are grooves in the angle of the inner and outer toes, after trimming, they are on the same horizontal line*



⑥ *After trimming, the inner and outer toes are on the same horizontal line, and the distance between the inner and outer toes at the posterior one third of the hooves is 5mm*



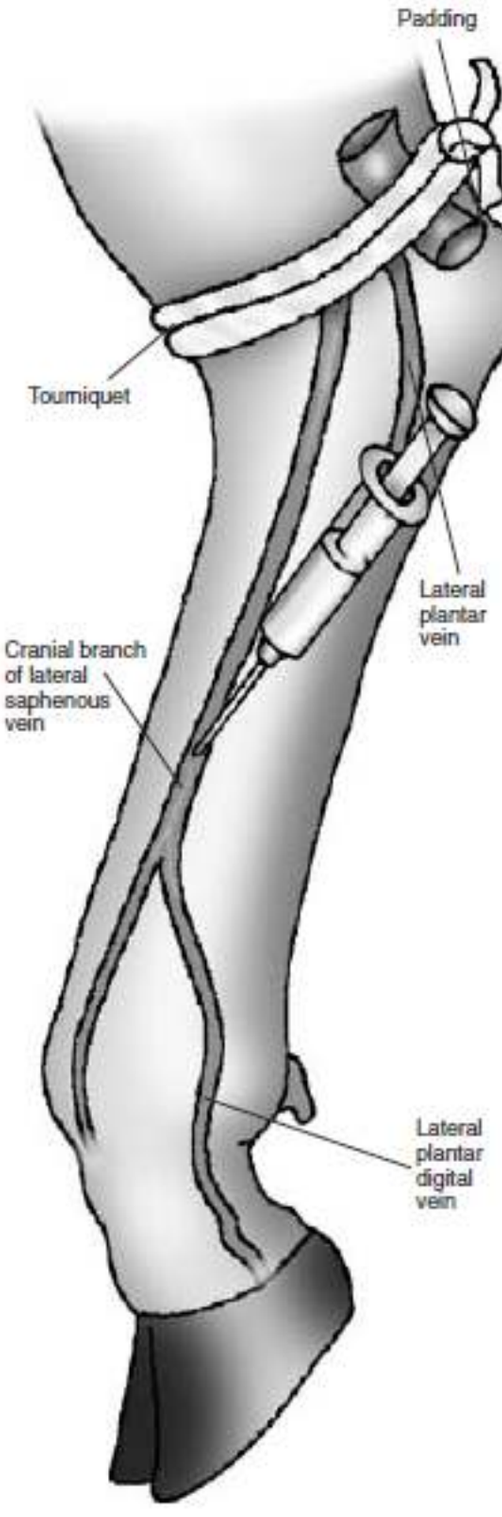
荷蘭五步驟修蹄法

- 一般性修蹄

1) 修剪蹄尖長度至正確長度: 蹄尖的長度，從冠狀帶量測 7.5 公分，厚度保留 0.5-0.7 公分，蹄的角度在 45-50 度之間
2) 外側蹄修整到跟內側蹄同長同高
3) 清理蹄底潰瘍部位(修整成淺碗型)

- 治療性修蹄

4) 減輕疼痛側蹄的負重
5) 移除病兆區域中鬆動/糜爛壞死組織



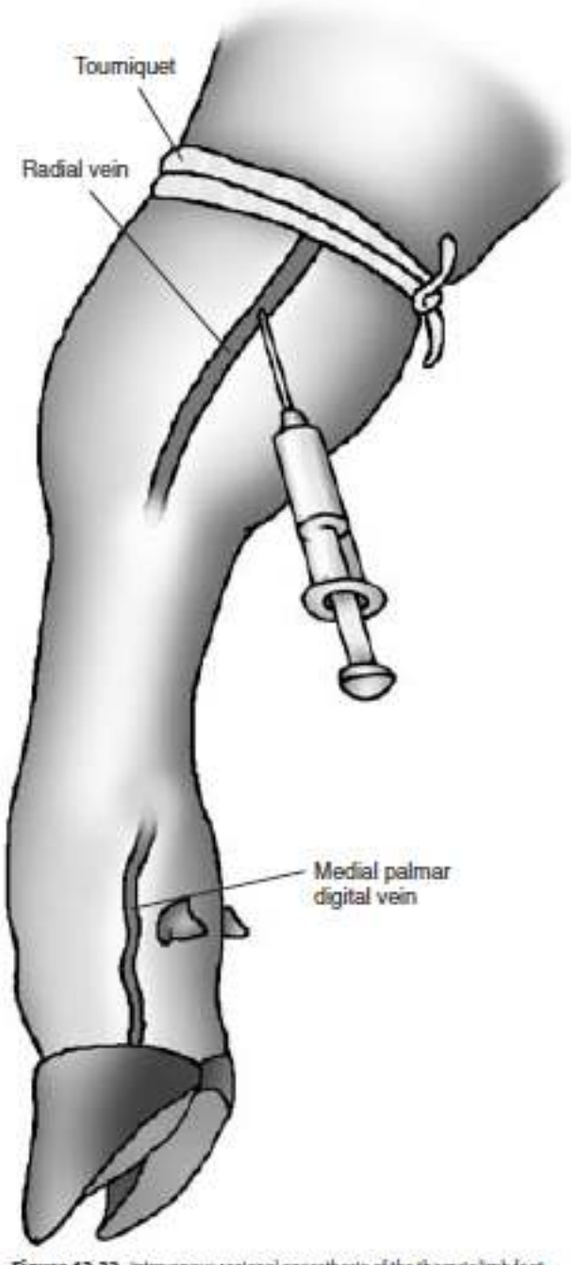


Figure 13.23 Intravenous regional anaesthesia of the thoracic limb foot (medial view).

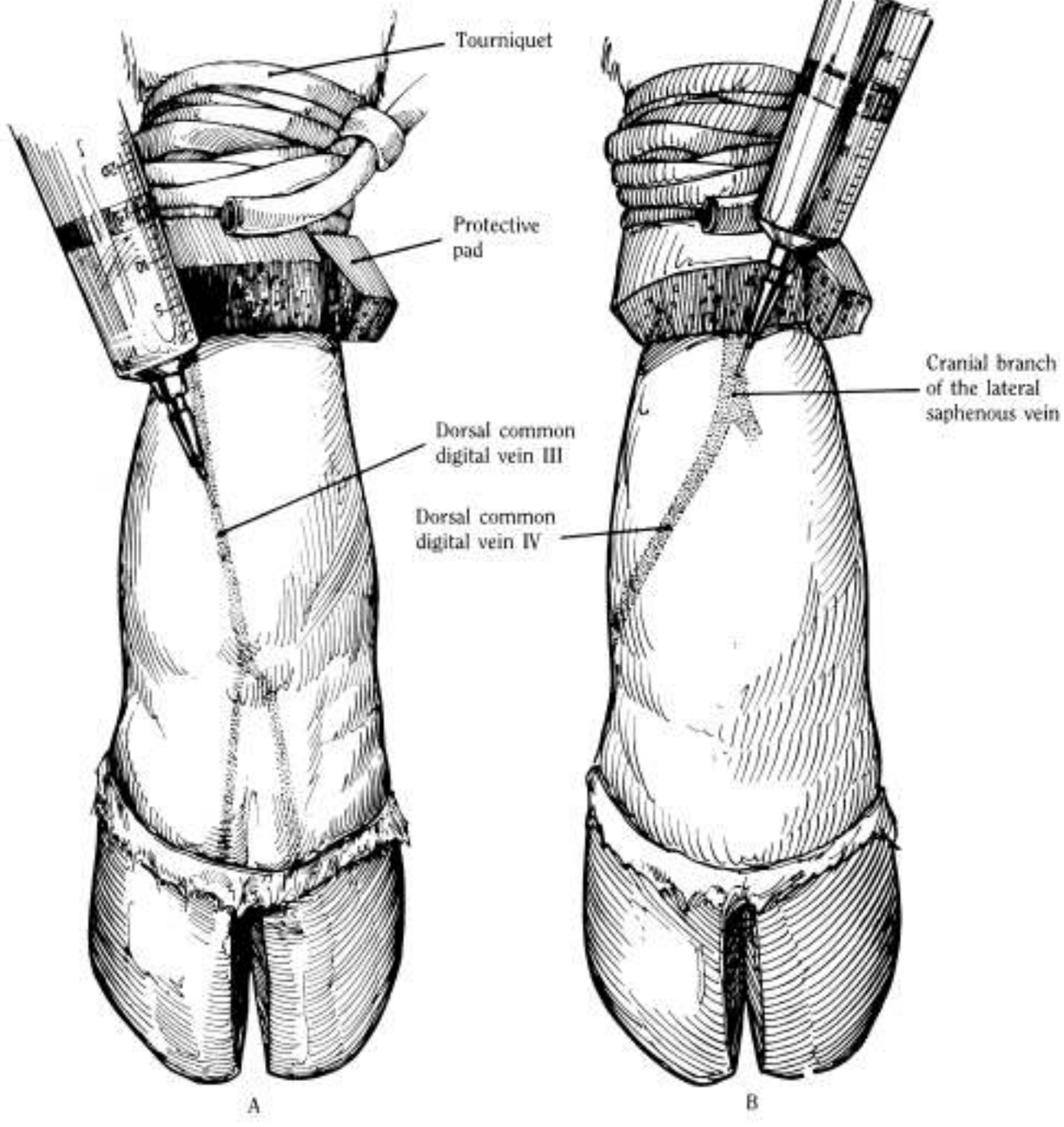


Fig. 2.5. Intravenous limb anesthesia. **A.** Forelimb, dorsal aspect. **B.** Hind limb, dorsal aspect.

Thank you

Questions?